



TRABALHO API SOAP

Disciplina: Desenvolvimento de Sistemas Distribuidos
Professor: Gracon Lima

PEDRO RICARDO E EDUARDO MEDEIROS

TECNOLOGIAS



PYTHON
Backend



JAVASCRIPT
Backend



HTML
Frontend

O PROBLEMA

Criar uma api usando o protocolo SOAP, com 2 endpoints no mínimo e com um cliente consumidor em tecnologia diferente da api.



IDEIA

Criar uma calculadora usando soap com uma api python, um cliente consumidor express.js e um front com html e css com javascript.

Calculadora SOAP

• OPERAÇÃO

VALOR A

5

VALOR B

3

Somar

Multiplicar

Dividir

Subtrair

5 + 3 = 8

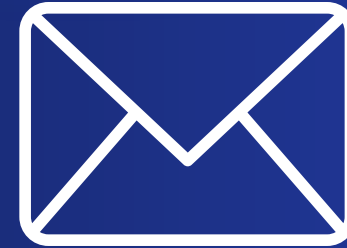
XML Enviado

XML Recebido

```
<?xml version='1.0' encoding='UTF-8'?>
  <soap11env:Envelope xmlns:soap11env="http://schemas.xmlsoap.org/soap/envelope/" xmlns:tns="calculadora.soap">
    <soap11env:Body>
      <tns:somarResponse>
        <tns:somarResult>8.0<class="xml-tag"></tns:somarResult>
      <class="xml-tag"></tns:somarResponse>
    <class="xml-tag"></soap11env:Body>
  <class="xml-tag"></soap11env:Envelope>
```

:8000 API SOAP Python → :3000 Cliente Node.js → :5500 Frontend HTML

SOAP



TESTE SOAPUI

The screenshot displays the SoapUI interface with a request and response view. The request is a SOAP message to a local service, and the response is a SOAP message returning a division result.

Request 1
URL: `http://localhost:8000/`

Raw XML

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <cal:dividir>
      <!--Optional:-->
      <cal:a>10</cal:a>
      <!--Optional:-->
      <cal:b>2</cal:b>
    </cal:dividir>
  </soapenv:Body>
</soapenv:Envelope>
```

Raw XML

```
<soapl1env:Envelope xmlns:soapl1env="http://schemas.xmlsoap.org/soap/envelope/">
  <soapl1env:Body>
    <tns:dividirResponse>
      <tns:dividirResult>5.0</tns:dividirResult>
    </tns:dividirResponse>
  </soapl1env:Body>
</soapl1env:Envelope>
```

Auth Headers (0) Attachments (0) WS-A WS-RM JMS Headers JMS Property (0)

response time: 26ms (287 bytes)

Headers (5) Attachments (0) SSL Info WSS (0) JMS (0)

1:1

TESTE SOAPUI

The screenshot displays the SoapUI application interface. The top toolbar includes icons for running tests, saving, and other functions. The address bar shows the URL `http://localhost:8000/`. The main workspace is split into two panels: the left panel shows the raw XML of the request, and the right panel shows the raw XML of the response.

Request XML:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <cal:dividir>
      <!--Optional:-->
      <cal:a>5</cal:a>
      <!--Optional:-->
      <cal:b>0</cal:b>
    </cal:dividir>
  </soapenv:Body>
</soapenv:Envelope>
```

Response XML:

```
<soapl1env:Envelope xmlns:soapl1env="http://schemas.xmlsoap.org/soap/envelope/">
  <soapl1env:Body>
    <soapl1env:Fault>
      <faultcode>soapl1env:Client</faultcode>
      <faultstring>Divisão por zero não é permitida.</faultstring>
      <faultactor/>
    </soapl1env:Fault>
  </soapl1env:Body>
</soapl1env:Envelope>
```

The bottom status bar provides additional details: the left side shows tabs for 'Auth', 'Headers (0)', 'Attachments (0)', 'WS-A', 'WS-RM', 'JMS Headers', and 'JMS Property (0)', with a message 'response time: 30ms (335 bytes)'. The right side shows tabs for 'Headers (5)', 'Attachments (0)', 'SSL Info', 'WSS (0)', and 'JMS (0)'. The system clock in the bottom right corner indicates the time is 11:20.



OBRIGADO!
